Structural models built from the drawings

31168 YMCA Langara and 31138 2170 W 1st · prepared 7 August 2026

Andrea's ETABS models were lost with a failed drive. Rather than re-enter the building's geometry by hand, this reads the concrete-outline DXF plans drafting already exports and produces an ETABS model covering every storey. It does none of the engineering — it removes the typing.

This one document covers both models. They are not equivalent: **31168 is the rebuild** — none of it existed. **31138 is a gap-fill**, because you had already rebuilt that building's walls by hand.

WHAT WAS GENERATED	31168 YMCA LANGARA	31138 2170 W 1ST
Storeys populated	**14**	29
Wall panels, true thicknesses	**294**	205
Columns, sized & oriented	**629**	304
Floor plates	**16**	15
Headers over openings (spandrel shells)	**17**	22
Shaft and stair openings cut	**44**	3
Pier labels	**126**	142
Members you already had, not duplicated	—	348 walls, 391 columns

What changed after your two calls

Everything below came from your feedback on 7 August and is already in the files.

From your notes of 12 August, already in: 31138 now goes to `P5`. Your model stopped at `P3` while drafting issues *LEVEL P4* and *LEVEL P5*, so two whole parkade floors were being read and placed nowhere. Both storeys are added below P3 at the height your parkade already uses, with the base dropped by the same total so nothing above them moved — and they bring 60 walls, 43 columns and 2 plates with them. Header bounds are now the 18–60" you specified.

One correction since the version of 9 August, found in checking rather than by you. A mezzanine reads as the level it sits above, so on 31168 both the *LEVEL 1* and the *LEVEL 1 MEZZ* sheet were building into *LEVEL 1 MEZZ*, and the ground floor of both towers stood empty — 45 walls and 67 columns a storey too high. On 31138 the same fault put a mezzanine part plan into L01. Both now build where they are drawn: 31168 gains A-LEVEL 1 and B-LEVEL 1, 31138 gains Mezz.

Also from 12 August: a sheet title may LIST its levels rather than range them. Three 31138 sheets are titled *LEVEL 8, 9*, *LEVEL 11, 12* and *LEVEL 17, 18*, and only the first number was ever read — so L09, L12 and L18 were built empty, and nothing said so, because each sheet reported itself placed on the one storey it did find. Listed titles are now read whole: 31138 goes from 26 populated storeys to 29.

Also from 13 August, and the largest single recovery yet: outlines were being stitched across any layers that shared a role, so a broken outline on one layer could capture the segments of a clean one on another. On 31168's *LEVEL 27 tower B* sheet, `JBP_V-WALL`'s six closed loops were welded to `JBP_B_WALL-1`'s twelve open chains and the core came out as a single 326×173 rectangle that resolves to nothing — one wall written where the drawing holds ten. Outlines now close per layer family first, and only what is left open is pooled, so a boundary Revit genuinely split across two layers still closes. **L27 goes from 1 wall to 18, matching every neighbouring storey**, and 31168 gains 22 walls and 5 headers overall.

Also 13 August, found by an independent audit and the most serious thing in this document: the DXF reader understood lines, arcs and polylines and had no case for `INSERT` at all. Anything drafting places as a *block* — which on 31138 is every steel column and every round concrete one — was therefore never read. Not dropped, not flagged: never read, so no count moved and nothing in the report said a word. **31138 places a hundred of its columns that way, and 75 were absent from the model**, whole floors at a time: all 22 on level 5, all 22 on level 6, all 15 on the mechanical level, all 15 on the roof. The reader now reads the block section and places each insert with its own scale and rotation, and the size limits carry the same half-inch of slack the wall rules do — the HSS 6×6 sections close at exactly 6.000000×6.000000 and were still falling a hair under a bare 6" minimum. All 100 are now modelled. On 31138 that is 304 columns where 244 stood, and 205 wall panels where 202 did. *31168 contains no blocks at all, so none of its numbers move.*

YOU SAID	WHAT IT DOES NOW
"We can't have a wall go from here to here and then another one from here to here. We need a connection."	Wall centrelines form a network. Corners are carried out to where they cross and share a joint; a wall running into another splits it so the T has a joint on both members.
"We don't model them as line elements, we model them as shells... in ETABS they're called spandrels."	Headers are wall panels standing only over their opening, depth = storey height less opening height. Both numbers now come off your own 31138 model rather than a standard: its 29 spandrels imply an opening of 86–88" (88" commonest, where 84" was assumed), and they run 20" to 60"; you then set the range yourself — 18" to 60". 365 of them on 31168, labelled as spandrels.
"Every floor we have two slabs on top of each other."	Drafting issues both a range sheet and a sheet per level, so each floor arrived twice. 43 doubled plates removed.
"Below grade, the basement walls are missing."	They are drawn as two concentric rings, so each read as a building-sized rectangle and both were discarded. Now read as the one wall they are — all four sides, including the angled west one.
"These are supposed to be round columns, but they're square. Square and rotated."	A column is now modelled round only where the drawing drew it with an arc, not where its shape merely looks circular — a chamfered square passes every shape test a circle does. 31168 comes out with exactly the three diameters its drawings contain — 16", 24" and 30" — and none besides.
"It filled the volume, but didn't do the actual corner."	Fixed, and it was a bug rather than a judgement call. An L now comes apart into its two limbs, each on its own centreline — which is what your own 31138 model says it should be: fifteen of its pier labels carry several panels on one storey, and three of those group limbs at right angles (cw9 a 23" limb with a 27", cw15 a 272" with a 104"). What stopped it was never a threshold. A solid-footprint test ran *before* the decomposer and claimed anything filling most of its bounding box, and an L fills 85% of its own; handed the same outline directly, the decomposer had always returned both limbs. It is asked first now, and the one-pier branch keeps only what genuinely does not come apart. Tower B's north-west corner had been a single panel 67 long and 42 thick — thicker than anything drawn there, the leg gone and the doorway under it gone with it. Across 31168 it recovered 84 more wall panels and 156 more headers, and the 55" opening on both corners.

"The lowest level, which is P3, seems way too high."	ETABS parks the base far below the building and folds the whole distance into the lowest storey — 31168 exported with the base at −1,000 ft and LEVEL P3 1,113 ft tall, so on import every member down there was extruded that far and the parkade arrived as a solid block half the height of the building. The storey list itself is now rewritten: the lowest storey takes a typical height from its own building and the base rises by exactly as much, so every other elevation is untouched. 31168 spans 104–561 ft, with the roof still at 561.1 ft and the lowest slab still at 113.9 ft.
"Less than 48 in length should be a column" — then "this should be a wall" for core faces	Connection decides, not length: under 48" and joined to no other wall is a column; a short face forming part of a core stays a wall.
Loads, diaphragms, stiffness modifiers, section properties are yours	Removed. No diaphragms are assigned, no loads applied.

What you give it

Two things, and neither is made specially for it.

YOU PROVIDE	WHAT IS READ FROM IT
The folder of plan DXFs drafting already exports for that job	Wall outlines, column footprints, slab edges, and the sheet title that says which level each drawing is.
Any ETABS file from the same job — `.e2k` or `.$et`, even an empty shell with nothing but the storeys	The storey list it builds into, the materials and sections it reuses, the grids, and anything you have already modelled, which it then leaves alone.

No configuration, no tagging, no naming anything up front. If you already have a partial model, that is the file to give it.

WHAT THOSE NEED TO CONTAIN	WHY
Walls on a `*WALL*` layer, columns on `*_COL*`, slab edges on `*SLABEDG*`	It is how a wall is told from a column from a floor. Layer names are configurable, but something has to separate them.
A sheet title naming its level — `LEVEL 29 PLAN (L29–35)`, `A-LEVEL 28`, `LEVEL P2`, `FOUNDATION`	It is how a drawing finds its storey. A range in the title fills every storey it covers.
A storey list and one concrete material in the ETABS file	It builds into your storeys at your elevations, never its own.
Both out of the same Revit project	They then already share an origin — on 31168 the core walls land on grids 15 and 16 to within a twentieth of an inch. Where they do not, you give it an offset once.

> **Two things decide how well it works, and both are drafting-side rather than engineering-side: layer discipline, and sheet titles that match the model's storey names.** Every placement fault found so far traces to the second — a model naming a storey `P1` where the sheet says `LEVEL P1`, or a `LEVEL 1` sheet colliding with `LEVEL 1 MEZZ`. The accuracy of this tool is a function of KOR's drafting standards, so tightening those improves every model it will ever build.

What it does not do

No section properties, no loads, no stiffness modifiers, no diaphragms, no design. That is your work and it is deliberately left untouched — a diaphragm arriving with the geometry is one more thing to undo. It also never overwrites what you have already modelled: on 31138 it recognised 348 of your walls and 391 of your columns and left every one of them alone.

What to open

Each sits in its project's `02 Engineering\02 Lateral Design\01 ETABS Models\`. In ETABS: **File** → **Import** → **ETABS .e2k Text File**.

FILE	WHAT IT IS
`31168-FROM-DRAWINGS.e2k`	**The one that matters.** The YMCA, its podium and the parkade — 14 storeys, 294 walls, 629 columns, 16 plates, every storey floored. One model per building, as you asked: nothing belonging to either tower is in this file. The ground floor, drafted twice as A-LEVEL 1 and B-LEVEL 1 1.7" apart, is one storey called LEVEL 1 — "it's all one big slab at L1, one elevation". Nothing of this existed before.
`31138-FROM-DRAWINGS.e2k`	A gap-fill. You have already rebuilt this building's walls — between 17 and 42 panels on every storey — so this adds only what the drawings show and your model does not: 205 walls and 304 columns, mostly at podium and parkade. 348 walls and 391 columns of yours were recognised at those locations and deliberately not modelled again — more than before, because a member now sits on the storey it runs to and so lands where yours already is.
`…-QUESTIONS.xlsx`	Three sheets. *Questions* — every judgement the tool had to make, each marked **DECIDED** with the measurement behind it. Nothing in it is waiting on you; rows tied to a rule can be changed from the answer cell. *Rules in force* — every rule the model was built on, read-only, including the geometry tolerances no question asks about. *Review & override* — every location where a decision was applied, each marked **APPROXIMATION** (a guess you can replace) or **DRAWING LIMIT** (the drawing lacks what would be needed).
`…-report.txt`	Not for reading end to end — it is the lookup for "why isn't this in here?". Every sheet, the storeys it landed on, what it produced, and every location that needed judgement.

Everything generated is named `KW*`, `KC*`, `KP*` with new sections prefixed `KOR-`, so you can select, filter or delete the whole lot in one action. Where the model already had a section of the right thickness it was reused, so members carry the project's own concrete and names you recognise.

What has been verified, and what has not

The four ways a model can be wrong, and which of them a count can see

Everything found in this programme has been one of four things, and only two of them show up in a total. A member built on the wrong storey leaves every count correct, which is exactly how an earlier version of 31168 was nearly sent with the ground floor of both towers empty. So the model is now checked against the *drawings*, in both directions, rather than against itself.

FAULT	SHOWS IN A COUNT?	WHAT CHECKS IT	WHERE IT STANDS
Geometry **dropped** — read and modelled as nothing	yes	Every drawn member must be modelled, or already be in your model at that place	36 of 4,229 unaccounted; may only ever come down
Geometry **doubled**	yes	Nothing may occupy the same place on the same storey — walls, columns, plates, headers	none
Geometry **misplaced** — right member, wrong storey	**no**	Every member must stand on linework from a sheet placed on its own storey; and no storey between two populated storeys may be empty	none
Geometry **misclassified** — right place, wrong size or shape	**no**	Built size and shape against the raw DXF entities, with roundness taken from the arc flag rather than from any judgement of shape	none

Each of those checks is proved against the defect it exists for before it is trusted. Undo the storey-matching fix and the audit answers `B-LEVEL 1, A-LEVEL 1 sit between populated storeys with nothing on them`. Undo round-column detection and it answers `KC217 drawn with straight lines only but built round`. A check that has never been seen to fail is not a check.

Verified: the model is drawn and looked at before it is sent

Counts and coordinates cannot see the faults that make a model wrong at a glance — a wall two inches tall passes every count. So the model is rendered to an image — isometric, two elevations, plan — and looked at before the file goes anywhere. Several faults were found that way and fixed, none of which any numeric check had noticed:

- **Joints carried elevations.** An ETABS joint is a plan position only — the third number on a `POINT` line is an offset from the storey the member is assigned to, and elevation comes from that storey. Writing a true elevation there throws a member hundreds of feet off its storey. Your own 31138 model settles it: 1,098 of its 1,156 joints carry nothing but X and Y. Members are now written the way ETABS writes them — a wall as its two plan joints repeated (`PANEL 4 "a" "b" "b" "a" 1 1 0 0`), a column as one joint rising a storey.
- **Towers were built as wafers.** 31168's storey list is the whole site, so it holds a storey for every tower's floor: because tower B's level 34 sits 2.4″ above tower A's, the export contains a storey named `B-LEVEL 34` that is two inches tall. Taking a storey's own height as its wall height made 78 panels two-inch wafers hanging a storey above their floor. A wall now stands on the previous floor *of its own tower*. The towers interleave at 4.67 ft and 5.0 ft, so no numeric threshold could have separated a real storey from a duplicate floor — only the storey name can.

  And it is one panel, not a stack of them. Assigning that wall to every storey it passed through did build it to the right height, but as several panels with a mesh break at each of the other tower's floor levels. The engineer drew the difference — "*when modelling the walls of tower B he should ignore tower A elevation system. The walls* *should not break at tower A elevations*" — so a member now carries a storey *span* instead: one panel from its own floor to its own ceiling, passing the other tower's levels without a joint. On 31168 that took wall panels under 6 ft from 390 to 63, and columns from 921 to 4. A member may only span past storeys belonging to another tower; spanning its own tower's floor is a fault and fails the build.

- **Scraps of floor hung in space.** Slab-edge linework closes into small rings that are not floors. Seven of 31138's fourteen plates were 52–68 ft^2 against a real tower floor of 9,666 ft^2. Measured across both projects the two populations do not overlap — standalone rings come out at 52–115 ft^2, real plates at 915 ft^2 and up — so a ring must now reach 400 ft^2 to be modelled as a plate.

Those three are now permanent checks, run against both real projects on every build: no wall or column shorter than a person, none taller than a double-height storey, no plate smaller than a room, and no storey in the reference that is not storey-height. The core regression suite is run before publishing.

Verified: every member sits exactly where the drawing draws it

Each plan is rendered with the extracted members drawn over it (`takeoff dxf-render <plan> <png> --overlay`), so what the model carries can be compared against the linework directly rather than inferred. On B-LEVEL 30 the core's two shafts, its bar wall and all twenty-four perimeter columns land on their outlines; the images are in `_drawing-renders\`. Wall thicknesses and column sizes are measured from the outlines, not assumed.

Verified: against your own model

On 31138 the generated columns land where you placed yours by hand: no shift applied, 43 of your 87 columns matched within six inches, and on L10 the median distance was 0.0. On 31168 the wall inventory agrees with the model that was lost — its pier-force export records five wall piers on every typical storey and 34 at L01, and the drawings give five core elements per typical storey and 29 walls at L01.

Verified: the geometry lands on your grid

The drawings and the model both come out of the same Revit project, so they already share a coordinate system and the geometry is placed without any shift. That is checked against the grid itself rather than assumed: the tower core straddles grids 15 and 16, which ETABS places at x = 1500 and 1826, and the walls read out of the drawing sit at 1500.05 and 1826.05 — a twentieth of an inch. The same holds up the tower, with wall coordinates landing on grid lines on B-LEVEL 30, B-LEVEL 33 and LEVEL 20.

Worth knowing when you look: only the inner face of a core wall sits on its grid line, the outer face standing off by the wall thickness, which is how the drawing draws it.

What it reads, and how

LAYER	BECOMES	HOW
`JBP_V-WALL`	Wall panels	A core is drawn as one ribbon tracing both faces of a group of walls. Each face is paired with the face opposite it; the pair gives the centreline and the true thickness.
`JBP_V_COL`	Columns	Footprint gives size and rotation, so a 16×24 sits the way it is drawn.
`JBP_C_SLABEDG*`	Floor plates	Outlines stitched into rings; rings inside a ring are openings.

Sheet titles carry the placement. `LEVEL 29 PLAN (L29-35)` fills seven storeys; `A-LEVEL 28` goes to Tower A alone; `LEVEL P2` to the parkade. The geometry is merged into an `.e2k` that ETABS itself exported, so storeys, grids, materials and design settings remain ETABS's own — the file differs from a known-good one only by the lines added.

Limitations — read before you rely on it

Slab plates — no longer the weak half, but still the part to check

Drafting interrupts slab edges wherever other linework crosses them, so many outlines do not close as vectors. On 31168 the slab-edge layer arrives as sixty-odd open chains, and at every tolerance from 0.05 to 72 inches the largest region they enclose is 119 ft² — so no tolerance was ever going to produce those floors, and widening one was measured and abandoned. They are now recovered by rasterising the drawn linework and flood-filling it, which is why **every storey carrying members now has a floor**. The 16 plates have a median area of 14,906 ft². LEVEL 1 has a floor for the first time — 78,859 ft² on a storey whose slab edge has never closed as vectors — recovered by flood-filling the drawn linework, and the report says so plate by plate.

The cost of that is the edges. A plate recovered from a raster is traced off pixels, so its outline comes out slightly stepped rather than crisp. The area and the position are right; the edge is approximate. Clean the outline wherever it matters.

Two other things worth knowing about the plates on this job. **C-LEVEL 3 has no closed slab edge of its own** and was given C-LEVEL 4's plate — the storey whose own floor is closest in shape to what stands on C-LEVEL 3 — and it is marked INFERRED in the report. And **LEVEL 2's outline closed through its own linework**, arriving as one ring of 296 × 96 ft whose edges met at a point; it is two podium wings and is now modelled as two plates of 12,380 and 12,271 ft².

Where no slab edge closes, the floor is taken from the inside face of the perimeter wall. You said a general outline would help to start from — "we can even have just one thickness per floor, general outline at first" — and on a below-grade level that wall is the only closed boundary the drawing offers. 31168's three parkade levels each get one plate of 75,800 ft², which is the site footprint. It is an approximation, offered because a storey with no plate has no diaphragm at all; replace it with your own outline wherever it matters.

Storeys carrying members with no plate: on 31168, none. Six had none when this was first written — `A-LEVEL 1`, `B-LEVEL 1`, `C-LEVEL 3`, `LEVEL 1 MEZZ` and two tower levels no longer in the model. Flood fill recovered four of them and C-LEVEL 3 borrows its neighbour's. Any storey that still had none would be named at the top of that project's report.

What the report does still name is the opposite case: a storey whose floor *reaches* less ground than the structure standing on it. On 31168 that is `LEVEL 2` at 43% and `LEVEL 1 MEZZ` at 4%, against 99–100% everywhere else. A mezzanine over part of a room and a podium that ends where a tower begins both look exactly like that and are correct; so does a slab edge that failed to close, and that is not. It is asked rather than guessed.

A floor with nothing under it, and members a second sheet lost

Two things the run now says out loud, because both were happening silently.

One storey on 31168 carries a floor plate with no wall or column beneath it: `C-ROOF`. A plate is held up by the members assigned to its own storey, so that is a roof slab supported by nothing. It is not a geometry error the tool can repair — the plan placed on that storey draws no vertical structure at all — so it is reported and left for you: either building C's structure genuinely stops below the roof, or the sheet that carries its walls is not the one we read. *This is a question, not a claim.*

2,145 members were drawn on a sheet in a place another sheet had already filled and were not added a second time. That is normal — a range plan and a per-level plan draw the same floor, and one of them has to give way — but until now the run discarded them without a word, so a sheet could report the members it contributed and put none of them in the model. The storeys that lost members that way are now named in the report. Tower A's level 35 plan is the case that found this: five walls and two columns against its name in the sheet table, none of them in the model.

Widening the closure tolerance was tried and rejected on the evidence: at 18″ the model gained 302 more plates, but the count of plates in a plausible size range did not change at all — every addition was a fragment. Recovering the missing plates needs a better method, not a looser one (see below), and that is why only one model is shipped.

What is set up, and what is yours to decide

The model arrives load-ready rather than load-less. Load patterns (Dead with self-weight, Live), load cases (Modal at 55 modes, Dead, Live) and the mass source come through from the reference, and because every wall, column and plate now carries its true thickness and the project's own concrete, **self-weight is real the moment you run it**. On 31138 your own 149 area-load assignments are preserved untouched.

What is not applied is superimposed dead and live load on the generated plates, and that is deliberate. Your own 31138 model carries five different SDL values on L01 alone — 5, 25, 45, 70 and 145 psf — with live load from −50 to 150, because they follow occupancy: terraces, planters, amenity, suites, corridors. None of that is visible in a structural outline, and putting a storey's "typical" value on a landscaped terrace would flow straight through to the foundation loads.

No diaphragm is assigned to a generated plate. That is deliberate — it goes on with the loads and modifiers, which are yours, and one arriving with the geometry is one more thing to undo. Also yours: spectra, stiffness modifiers, and meshing.

Shaft and stair openings are cut

Inner rings on the slab-edge layers are cut out of the plate they sit in, written the way ETABS marks an opening — an area carrying no section. The counts are in the table above.

> **Check the thick walls.** Anything modelled thicker than 24″ is flagged in the report. Wall outlines that meet at junctions can pair across the junction and read thicker than the wall really is. The flags tell you exactly where to look.

31168's storeys are the whole site, not Tower B alone

The model that was lost was Tower B on its own, storeys L01 to L40 with Mezz and P1–P2. This one is built on the site model exported from Revit, so its storeys are the whole development — `B-LEVEL 27` upward for the tower, shared unprefixed storeys lower down. The geometry is right; the frame it sits in is not the one you were working in. Worth deciding before you build on it.

It believes the drawing

Thicknesses and sizes come from what was drawn, not from a schedule. A wall drawn at the wrong width models at the wrong width. The plans used here are dated 28 June 2026 — later revisions are not in this model.

The decisions, and everything open to override

Beside each model is `…-QUESTIONS.xlsx`, in three parts.

Questions lists every judgement the tool had to make without being told — and every one has been taken. Each row carries what was decided and the measurement behind it, so you can disagree with a specific number rather than with the tool. You are not asked to confirm any of them; nothing here blocks you from importing the model.

Two kinds of row, and the difference matters. **DECIDED** rows are tied to a rule: write in the answer column and the next model is built your way, on this job and every job after. **SCOPE** rows record what the tool does or does not attempt at all — beams, for one — and there is no number behind them to change. An answer to a SCOPE row is recorded and read, but it will not move geometry by itself, so those are worth saying to us directly. The status column says which is which, and the answer box is only offered on the rows that can act on one.

Rules in force is the whole rule set the model was built on, read-only: the value, where it came from, and why it holds. Most of them have a row on the first sheet and it says which. The ones that do not are geometry-cleanup tolerances — how far a dashed outline may be bridged to close, how close a generated member must sit to one of yours to count as the same member. Nothing there is a question, but if an outline did not close and you want to say why, that page is the number to name.

Review & override is everything else it decided for you, grouped, counted, and each marked one of two ways. **APPROXIMATION** is a defensible guess you can replace — a floor taken from the inside of a perimeter wall, a short element made a column, the lowest storey given a typical height. **DRAWING LIMIT** means the drawing does not contain what would be needed: an outline that will not close cannot become a plate, whatever the tool does. The first kind is a choice; the second takes you or a redraw. Nothing in that list is an error — errors are fixed rather than listed.

Those answers are the point. Each one becomes a rule the tool applies from then on, on this project and every one after, instead of a judgement folded into the code where nobody can see it. The workbook also carries every flagged location and a ledger of which drawing filled which storey, so a wrong answer is traceable to the sheet that caused it.

How it gets better — and why that depends on you

Everything this tool knows, an engineer told it, and every rule in it can name where it came from. That is not a figure of speech: the rules live in a database with the evidence and a confidence level attached, and they are applied only once they reach *replay-verified* or *engineer-confirmed*.

WHAT YOU DO	WHAT HAPPENS TO IT
Override a decision in the workbook.	Your answer becomes a rule at *engineer-confirmed* — the highest confidence there is — and applies to every model built from then on, on every job. It supersedes ours permanently; you are asked once.
Say something is wrong when you open a model.	It is measured against your own models and the 400-odd on the share, and becomes either a rule or a question with the measurement attached. Nothing is changed on a hunch.
Suggest something it should do that it does not.	That is how the last four capabilities arrived — headers as spandrels, pier labels, openings cut, the basement wall. Every one started as a sentence on a call.

It already learns without asking. Three of the five questions you were going to be sent were answered by reading your own 31138 gravity model instead — and every one of them is now a decision in the workbook rather than a question in your inbox:

- **Header depth.** Taking each of your 29 spandrels' depth off its storey height gives the opening it was drawn for — 86" eleven times and 88" twelve. A standard 84" door had been assumed, so every header was two to four inches too deep. Fixed, and not asked.
- **Short elements.** You keep standalone 30" and 36" panels as walls with pier labels, and your most slender column is 12x36 — exactly 3:1, with nothing beyond it. So anything longer than three times its width is now a wall whatever it touches. Fixed, and not asked.
- **Corners.** Fifteen of your pier labels cover several panels on one storey and three group limbs at right angles, which says what a stepped block should look like. Built, and not asked: an L now comes apart into its two limbs on their own centrelines. What had been stopping it was order — a solid-footprint test ran before the

decomposer and claimed anything filling most of its bounding box, and an L fills 85% of its own. Tower B's north-west corner had been one panel 67 long and 42 thick, thicker than anything drawn there.

The more of your work it can read, the fewer questions you get. That is the whole design.

What would make it materially better — your input decides

IMPROVEMENT	WHAT IT NEEDS FROM YOU
Close slab edges properly by extending an interrupted edge along its own direction to the line that cut it, instead of bridging by distance. Measurement says this is the only way to recover the missing plates.	Nothing — but tell us which storeys you most need plates on, so it is proven where it matters.
Wall grades by height. Read the wall schedule so 65 MPa low and 45 MPa high come through instead of one material.	Point us at the schedule sheet you would read for this, and how you would expect the grades keyed to walls.
Beams. Nothing on the beam layers is read today.	Whether beams are worth having, or you would rather add them yourself.
A button in the KOR app instead of a command line.	Whether you would run this yourself per project, or ask for it.

The single most useful thing you can send back: open the model, and tell us the first three things that are wrong or annoying. Each one is a rule the tool does not yet know.

Where this sits in the KOR toolset

This is a new front end on the engineering engine the app already carries, not a separate program.

```
Revit  ──→  drafting's DXF plan exports  ──→  [ DXF → ETABS ]  ──→  .e2k  ──→  ETABS
PDF drawings ─────────────────────────────→  [ PdfToSafe   ]  ──→  .f2k / .e2k / live OAPI
PDF drawings ─────────────────────────────→  [ Takeoff     ]  ──→  quantities, xlsx
```

All three live in `Kor.Operations.EngineeringTools`, share the same geometry code, and are covered by the same test suite. PdfToSafe reads drawings when only a PDF exists; this reads the DXFs when drafting has exported them, which is exact and needs no vision model. The natural next step is a button in the app's Engineering Tools panel so it runs without a command line.

> **It only stays reliable while the layer names hold.** `JBP_V-WALL`, `JBP_V_COL` and `JBP_C_SLABEDG` are now a machine contract, not just a drafting habit — which is a concrete argument for the Revit template and standards work already under way.

Built 6 August 2026 · `Kor.Operations.EngineeringTools.Core/Dxf` · CLI `takeoff dxf-to-etabs`, `dxf-inspect`, `e2k-compare` · documented at `docs/DxfToEtabs.md`